Beyond Tokens per Second: Benchmarking LLM Serving Engines Under Load

Anonymous Author(s)

ABSTRACT

Large language model (LLM) services are deployed on open-source serving engines whose schedulers decide, under load, which requests run, which wait, and which are evicted. These decisions are invisible to the application, yet they determine what users experience. Published comparisons of serving engines mostly report peak throughput or per-token speed, which says little about how an engine behaves once its memory or scheduling capacity is exhausted. We present an empirical comparison of three widely used engines, vLLM, SGLang, and TensorRT-LLM, driven through their own OpenAI-compatible frontends by a common open-loop benchmark harness. We measure six user-facing metrics (time to first token, time between tokens, time per output token, end-to-end latency, throughput, and request outcome) and combine them into SLO goodput, across three models with different key–value (KV) cache footprints, realistic mixed and multi-turn workloads, five arrival processes, two scheduler capacity settings, two prefill-handling features, n-gram speculative decoding, and two hardware platforms.

The engines are nearly indistinguishable at light load and diverge sharply once a resource saturates, and they diverge in *where* they make requests wait. On heterogeneous traffic the engine with the fastest token generation delivers the lowest goodput, while on uniform traffic it is the fastest end to end; a performance feature that helps one engine can hurt another. We also report methodology pitfalls that change conclusions: time to first token measured from dispatch hides client-side queueing, finite traces make latency non-monotonic in load, and identically named engine metrics are not comparable. We conclude with a decision guide for practitioners and release the harness and data. [TODO: tighten to the final RQ findings; ICPE abstracts are typically 150–250 words]

CCS CONCEPTS

• **Software and its engineering** → **Software performance**; • **General and reference** → *Experimentation*; • **Computing methodologies** → Artificial intelligence.

KEYWORDS

LLM inference serving, benchmarking, vLLM, SGLang, TensorRT-LLM, KV cache, scheduling, SLO goodput, tail latency

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1 INTRODUCTION

Serving a large language model (LLM) is a queueing problem with an unusual resource: the key–value (KV) cache, which grows with every token a request generates and must stay in GPU memory until the request finishes. Modern serving engines—vLLM [12], SGLang [31], and TensorRT-LLM [19]—batch requests continuously at iteration granularity [30] and manage the KV cache in blocks, which lets them keep hundreds of requests in flight on one GPU. When the KV cache or the scheduler's slots run out, however, each engine must choose: admit a new request and risk evicting a running one, or hold the new request back. These choices are made inside the engine, are rarely documented as part of the engine's contract, and are not exposed through the OpenAI-compatible APIs that applications use.

Practitioners choosing an engine typically rely on published per-token throughput figures or on vendor benchmarks run at a fixed, comfortable load. Such figures answer the question "how fast can this engine generate tokens?" but not the question operators actually face: "what will my users experience when traffic is heavy, request lengths vary, and memory is tight?" The answer depends on several metrics at once—how long a user waits for the first token, whether the stream stalls midway, how long the whole answer takes, how many requests fail—and on which of these the application's service-level objective (SLO) is bound by.

This paper compares vLLM, SGLang, and TensorRT-LLM empirically under conditions that expose their scheduling behaviour. We drive all three through their own OpenAI-compatible serving frontends with a common open-loop harness, AnonBench, using realistic workloads assembled from public corpora, Poisson and bursty arrival processes, and three models whose per-token KV footprint differs by up to 6.25×, including one model with full multi-head attention (MHA) and two with grouped-query attention (GQA) [5]. We organise the study around four research questions:

- **RQ1** *Engine behaviour under load.* How do the engines compare in responsiveness, streaming quality, end-to-end latency, throughput, reliability, and SLO goodput as load and KV-cache pressure increase, and what scheduler behaviour explains the differences?
- **RQ2** *Prefill handling.* How do prefix caching and chunked prefill change latency under load on each engine?
- **RQ3** *Speculative decoding.* Does n-gram speculative decoding improve latency and throughput, and is the effect consistent across engines?

RQ4 *Hardware generalisation.* Do the **RQ1** findings hold on a different hardware platform with a much larger, unified memory?

Findings. [TODO: finalise against Section 5; the statements below reflect the current analysis]

- At light load the engines are interchangeable: decode speed differs by at most 6%. They diverge once a resource saturates, and the dominant difference is *where* each engine makes requests wait: vLLM admits early and later stalls running streams, SGLang holds requests before their first token and then streams smoothly, and TensorRT-LLM admits the fewest requests.
- The engine with the fastest token generation (TensorRT-LLM) has the lowest SLO goodput on heterogeneous traffic, yet the lowest end-to-end latency on uniform multi-turn traffic. Under the SLOs we study, goodput is decided by time to first token, not by token speed.
- Raising the scheduler's sequence limit doubles goodput when slots are the bottleneck and has no effect when the KV cache or the engine's admission policy is.
- Performance features are engine-specific: prefix caching helps all engines alike, whereas chunked prefill and n-gram speculative decoding help one engine and hurt another.
- The engine ranking under KV pressure is preserved on a second platform with 2.7× the memory and newer engine versions.

Contributions.

(1) AnonBench, a black-box benchmark for LLM serving engines with a formal definition of experiments, arrival-based latency metrics, censoring of timed-out requests, and SLO goodput. Engines, models, workloads, and arrival processes are parameters, so practitioners can evaluate their own deployment (Section 3).
(2) A controlled, multi-metric comparison of three production LLM serving engines under load, covering latency, streaming quality, throughput, reliability, and SLO goodput (Section 5).
(3) An explanation of the observed differences in terms of each engine's admission and eviction behaviour, the scheduler's sequence limit, and the KV-cache footprint of the model.
(4) A set of methodology lessons for benchmarking LLM serving, each of which changes a conclusion if ignored (Section 6).
(5) A decision guide for practitioners, and an open benchmark harness with all per-request data (Section 9).

The remainder of the paper is organised as follows. Section 2 summarises the serving mechanisms that matter for our results. Section 3 defines the benchmark and its metrics, and Section 4 instantiates it for this study. Section 5 answers the research questions. Section 6 discusses implications for practitioners and for benchmarking, Section 7 threats to validity, and Section 8 related work.

2 BACKGROUND

Prefill, decode, and the KV cache. An autoregressive transformer [28] serves a request in two phases. The *prefill* phase processes the whole prompt in one forward pass and produces the first output token; the *decode* phase then generates one token per forward pass. Each layer stores the keys and values of every processed token in the KV cache, so a request's cache footprint grows linearly with its prompt and output length. The footprint per token is $2 \times L \times H_{kv} \times d_{head} \times b$ bytes for L layers, H_{kv} KV heads of dimension d_{head}, and b bytes per element. Multi-query [25] and grouped-query attention [5] reduce H_{kv} below the number of query heads; models with full multi-head attention (MHA) keep one KV head per query head and therefore need several times more cache per token.

Continuous batching and paged KV memory. Serving engines batch at iteration granularity: requests join and leave the running batch between forward passes [30]. To avoid fragmentation, vLLM's PagedAttention stores the KV cache in fixed-size blocks allocated on demand [12]; SGLang and TensorRT-LLM use comparable block-based managers. The number of requests that can run concurrently is bounded by two limits: a configured maximum number of running sequences (`max_num_seqs` in vLLM; analogous settings in the other engines) and the free capacity of the KV-cache pool.

Admission, preemption, and eviction. When the KV pool fills, an engine must either stop admitting requests or reclaim memory from running ones. Engines differ in how much memory they reserve when they admit a request. Allocating on demand admits the most requests but may later run out of memory mid-generation, forcing the engine to *preempt* a running sequence, discard its cache, and recompute it later. Reserving memory for a request's maximum output length up front avoids eviction but admits fewer requests. [TODO: document each engine's default policy and the configuration used here: vLLM V1 preempt-and-recompute; SGLang retraction with an estimated new-token ratio; TensorRT-LLM capacity-scheduler policy (verify against the launch log)]

Prefill-handling features. Prefix caching retains the KV cache of previously processed prompts so that a request sharing a prefix—for example, the next turn of a conversation—skips recomputing it; SGLang organises this cache as a radix tree [31]. *Chunked prefill* splits a long prompt into fixed-size chunks that are interleaved with decode steps, bounding how long a long prompt can stall the running batch [4]. Disaggregating prefill and decode onto different devices [20, 32] is out of scope for this single-GPU study.

Speculative decoding. Speculative decoding drafts several tokens cheaply and verifies them in one forward pass of the target model, accepting the longest correct prefix [9, 13]. *N-gram* (prompt-lookup) speculation drafts by matching the most recent tokens against earlier text in the context and needs no draft model [23]. Its benefit depends on the draft acceptance rate and on the extra work verification adds to each batch.

User-facing latency metrics. Time to first token (TTFT) is dominated by queueing and prefill; the time between streamed tokens (TBT) and the time per output token (TPOT) by decode; end-to-end

latency (E2E) by both. Because no single metric captures user experience, recent work evaluates serving systems by *goodput*: the rate or share of requests that meet all SLO targets at once [32].

3 BENCHMARK METHODOLOGY

AnonBench is a black-box benchmark for LLM serving engines. It is defined independently of any particular engine, model, workload, or hardware: each is a parameter of an experiment, so practitioners can evaluate the deployment they care about. This section formalises the benchmark; Section 4 instantiates it for this study.

Figure 1 shows the four stages of the benchmark. An *experiment* is the tuple

$$\mathcal{E} = (\mathcal{W}, \mathcal{A}, C, M, e, \theta, h), \tag{1}$$

of a workload $\mathcal{W}$, an arrival process $\mathcal{A}$ with client concurrency limit C, a model M, a serving engine e with scheduler configuration θ, and a hardware platform h. The first five define the benchmark configuration (Section 3.1); e is the system under test (Section 3.2); the harness (Section 3.3) and the analysis (Section 3.4) are the same for every experiment.

3.1 Benchmark configuration

Workloads. A request $r_i = (\mathbf{x}_i, m_i, \kappa_i)$ consists of a prompt $\mathbf{x}_i$ of $p_i = |\mathbf{x}_i|$ tokens, a maximum output length m_i, and a task label κ_i. A workload is a mixture of K task distributions,

$$\mathcal{W} = \sum_{k=1}^{K} \pi_k \, \mathcal{D}_k, \qquad \sum_k \pi_k = 1, \tag{2}$$

where $\mathcal{D}_k$ draws prompts and output budgets from a corpus for task k. A *conversational* workload groups requests into conversations c with turns $r_{c,1}, \ldots, r_{c,J_c}$, where the prompt of turn j contains the full history, so $\mathbf{x}_{c,j-1}$ is a prefix of $\mathbf{x}_{c,j}$. Mixtures with heterogeneous p_i and m_i stress scheduling, conversational workloads stress prefix reuse, and long-prompt workloads stress prefill.

Load generation. The arrival process $\mathcal{A}$ assigns every request an arrival time a_i before the run, independently of the system's responses (open loop [24]). We use two families: a Poisson process of rate λ, with $a_{i+1} - a_i \sim \mathrm{Exp}(\lambda)$, and a bursty process that releases B requests, δ apart, every T_B seconds; any point process can be substituted. The client keeps at most C requests in flight, so request i is *dispatched* at

$$s_i = \inf\{\, t \geq a_i : n_{\text{inflight}}(t) < C \,\}, \tag{3}$$

and $q_i = s_i - a_i \geq 0$ is the time it waits at the client. Within a conversation, turn $j > 1$ is released only after turn $j-1$ finishes at $f_{c,j-1}$, i.e., $s_{c,j} \geq f_{c,j-1}$, as a user waits for an answer; conversations themselves arrive open-loop.

Models. A model enters the benchmark through its weights, tokenizer, and chat template, and through its key–value (KV) footprint. With L layers, H_{kv} KV heads of dimension d, and b bytes per element, each token occupies $2LH_{kv}db$ bytes of KV cache, so β bytes of cache memory hold

$$K = \left\lfloor \frac{\beta}{2\,L\,H_{kv}\,d\,b} \right\rfloor \text{ tokens.} \tag{4}$$

A running request occupies $p_i + g_i(t)$ of these after generating $g_i(t)$ tokens. Models with grouped-query attention (small H_{kv}) leave far more room than models with full multi-head attention.

Scheduler configuration. The engine configuration $\theta = (S, \phi)$ comprises the scheduler's limit S on concurrently running sequences and a set of optional features

$$\phi \subseteq \{\textsc{prefix},\ \textsc{chunk}(c),\ \textsc{spec}(k)\}\ : \tag{5}$$

prefix caching, chunked prefill with chunk size c, and speculative decoding with k draft tokens. All other options keep the engine's defaults and are recorded with each run. Concurrency is therefore bounded by S and by K. By Little's law [14], the mean number of requests in the system is $\bar{N} = \lambda \bar{R}$ for mean residence time $\bar{R}$: the scheduler limit binds when $\lambda \bar{R} > S$, and the KV pool binds when the tokens of running requests reach K. Which limit binds first, and what an engine does when it binds, is what the benchmark exposes.

Experiment orchestrator. The orchestrator expands a configuration grid into experiments (Equation (1)), launches a fresh server for each so that caches and counters start empty, waits until the server reports ready, replays the trace, collects the measurements, and shuts the server down before the next experiment. Every experiment's full configuration and the engine's launch log are stored with its results.

3.2 Serving engines as black boxes

Engines are driven exclusively through their OpenAI-compatible `/v1/completions` endpoint with token streaming. Each engine is therefore measured as applications use it, including its HTTP frontend, tokenizer, and scheduler, and a new engine is added with a thin adapter that only maps the configuration θ to the engine's launch options. The benchmark makes no assumption about an engine's internal design; differences in scheduling and memory management are inferred from client-side measurements and, where available, explained with the engine's own metrics.

3.3 Evaluation and monitoring harness

Metrics. For each request the client records a_i, s_i, the arrival times $t_{i,1} < \cdots < t_{i,n_i}$ of its n_i output tokens, and an outcome $o_i \in \{\textsc{fin}, \textsc{zero}, \textsc{to}\}$ (finished, ended without a token, timed out). The user-facing metrics are:

$$\mathrm{TTFT}_i = t_{i,1} - a_i, \tag{6}$$

$$\mathrm{TBT}_{i,k} = t_{i,k+1} - t_{i,k}, \qquad G_i = \max_k \mathrm{TBT}_{i,k}, \tag{7}$$

$$\mathrm{TPOT}_i = \frac{t_{i,n_i} - t_{i,1}}{n_i - 1}, \tag{8}$$

$$\mathrm{E2E}_i = t_{i,n_i} - a_i, \tag{9}$$

$$X = \frac{\sum_{i:\, o_i = \textsc{fin}} n_i}{\max_i t_{i,n_i} - \min_i a_i}. \tag{10}$$

TTFT is the time to first token, TBT the time between streamed chunks with G_i the longest stall a user sees, TPOT the time per output token, E2E the end-to-end latency, and X the output throughput. TPOT, unlike TBT, does not depend on how many tokens a chunk carries, which keeps it comparable when speculative decoding emits several tokens at once.

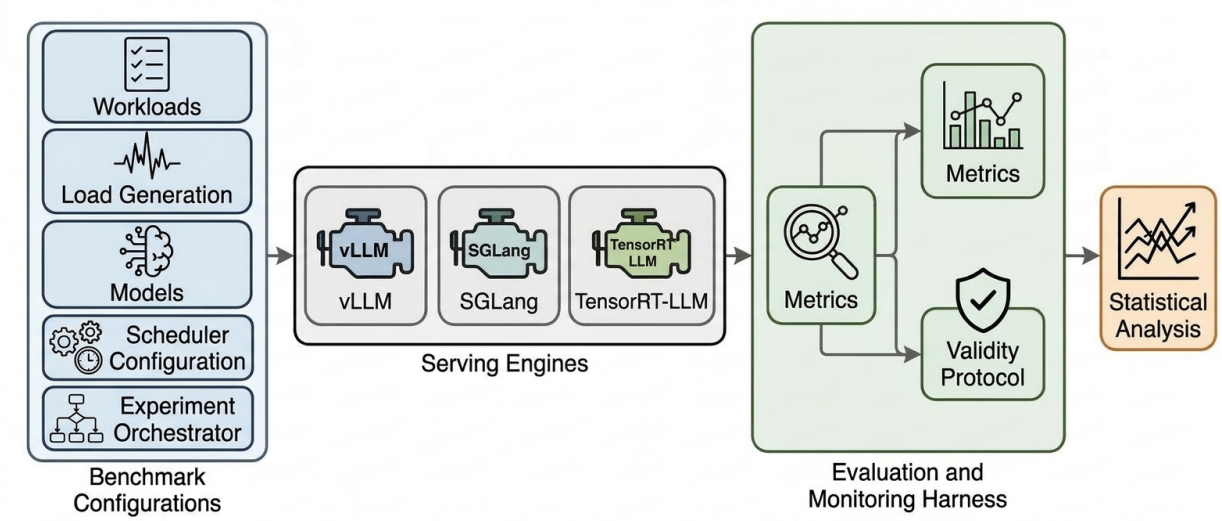


Figure 1: Overview of AnonBench. A benchmark configuration (workload, load generation, model, scheduler configuration) is executed by the experiment orchestrator against each serving engine. The evaluation and monitoring harness measures every request from the client side, polls the engine's own metrics, and applies a validity protocol; the statistical analysis derives the reported distributions, SLO goodput, and feature effects.

Engine-side monitoring. In parallel, the harness polls the engine's native metrics endpoint every Δt seconds for KV-cache occupancy, running and waiting sequences, preemptions, and prefix-cache hit rate. These signals have engine-specific semantics—cumulative versus per-batch windows, or availability only after a run—so they are used solely to *explain* client-side results, never to rank engines.

Validity protocol. Five rules keep the measurements comparable across engines.

(1) *Open-loop arrivals.* The trace is fixed before the run, so a slow engine cannot slow down its own offered load.
(2) *Arrival-based latency.* TTFT and E2E are measured from arrival a_i, so client-side queueing q_i counts against the system; measuring from dispatch would hide it (coordinated omission [27]). The exception is a later conversation turn, whose wait for the previous answer is the user's own; there TTFT is measured from s_i.
(3) *Censoring instead of dropping.* A request without a first token within τ_1 of dispatch is assigned $\mathrm{TTFT}_i = q_i + \tau_1$ and $\mathrm{E2E}_i \geq q_i + \tau_1$; one that does not finish within τ_2 is assigned $\mathrm{E2E}_i = q_i + \tau_2$. Dropping them would remove an overloaded engine's worst latencies.
(4) *Separating model failures from engine failures.* Zero-token replies reflect the model or prompt, not latency; they are excluded from latency statistics but count as goodput misses. Requests the client cannot send (e.g., prompts longer than the context) are excluded and reported.
(5) *Isolation.* Each experiment runs on a freshly started server with no other load.

3.4 Statistical analysis

Distributions. Per-request metrics are summarised by empirical quantiles $Q_p(\cdot)$, typically $p \in \{0.5, 0.95, 0.99\}$, and by empirical CDFs where the shape matters. [TODO: add bootstrap confidence intervals over requests for the reported quantiles]

SLO goodput. Given a TTFT target $T^\star$ and a TPOT target $P^\star$, goodput is the share of *all* requests that finish within both:

$$\Gamma(T^\star, P^\star) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\left[o_i = \text{FIN} \ \wedge \ \mathrm{TTFT}_i \leq T^\star \ \wedge \ \mathrm{TPOT}_i \leq P^\star \right]. \tag{11}$$

Goodput rewards neither speed on a subset of requests nor success with unusable latency, and it is the measure on which engines are ranked.

Feature effects. The effect of a feature set ϕ on a metric μ is reported relative to the same experiment without it,

$$\Delta_\mu(\phi) = \frac{\mu(\theta_\phi) - \mu(\theta_\varnothing)}{\mu(\theta_\varnothing)}, \tag{12}$$

so that features can be compared across engines whose absolute performance differs.

Load response. Metrics are reported as functions of the arrival rate λ and, to expose how a backlog forms within a run, of a request's position in the trace; the position at which TTFT first exceeds a threshold marks the point where a resource saturates.

4 EXPERIMENTAL SETUP

This section instantiates the benchmark of Section 3 for the four research questions.

Table 1: Engines and platforms.

	L40S (**RQ1–RQ3**)	GB10 (**RQ4**)
vLLM (V1)	0.11.0	0.32.0
SGLang	[VERIFY: version]	0.5.22
TensorRT-LLM (PyTorch)	1.0.0	1.2.1
Memory	44.4 GiB	121 GiB unified
KV memory fraction	0.90 [VERIFY:]	0.80

Table 2: Models. KV bytes per token at fp16.

Model	Attn.	H_{kv}	L	KV B/token
Llama-3.1-8B-Instruct	GQA	8	32	131,072
Mistral-NeMo-12B-Instr. [18]	GQA	8	40	163,840
Qwen1.5-14B-Chat	MHA	40	40	819,200

4.1 Engines and platforms

Table 1 lists the engines and platforms. All engines serve fp16 weights on one GPU (no tensor parallelism) with a maximum context of 8,192 tokens. The scheduler limit S maps to each engine's equivalent setting. [TODO: list the per-engine flags: KV memory fraction, maximum batched tokens, maximum running requests, maximum batch size, and the capacity-scheduler policy] Two engine-specific adjustments were necessary. TensorRT-LLM treats the chunk size as a hard prompt-length limit unless chunked prefill is enabled explicitly, so the chunked-prefill configuration sets that flag. TensorRT-LLM's n-gram speculation is incompatible with its overlap scheduler, which we disable whenever speculation is on; SGLang disables its equivalent optimisation automatically and caps running requests at 48 under speculation.

4.2 Models

We choose three instruction-tuned models whose per-token KV footprint $2LH_{kv}db$ (Equation (4)) spans a wide range (Table 2). Qwen1.5-14B-Chat [8] has no KV-head reduction ($H_{kv} = 40$) and needs 5× the cache per token of Mistral-NeMo-12B and 6.25× that of Llama-3.1-8B [16]. We considered distilled reasoning models but rejected them after confirming from their configuration files that they use GQA backbones, which would not provide the contrast.

Prompt formatting. The client sends plain instruction text to the completions endpoint. Qwen1.5-14B-Chat emits its end-of-turn token immediately on unformatted prompts, so for this model the client applies the model's chat template at send time. [TODO: decide whether to template all models; see Section 7]

4.3 Workloads

We instantiate three workloads from public corpora (Table 3). `mixed_real` is a mixture (Equation (2)) of four tasks with $\pi_k = 1/4$: chat (ShareGPT [1]), code generation (MBPP [7]), reasoning (GSM8K [10]), and summarisation (CNN/DailyMail [11]). It pairs short prompts with very long output budgets (reasoning) and long prompts with short outputs (summarisation), so a few long-running requests can hold scheduler slots and KV memory for minutes.

Table 3: Workloads per run (token counts are medians).

Workload	Task (source)	Req.	p_i	m_i
`mixed_real`	chat (ShareGPT)	125	20	239
	code (MBPP)	125	53	45
	reasoning (GSM8K)	125	85	6,810
	summ. (CNN/DM)	125	791	64
`chat_multiturn`	2–6 turns (ShareGPT)	280–600	454	310
`chat_real_long`	≥500 tok. (ShareGPT)	300	938	[VERIFY:]

Table 4: Experimental design.

RQ	Configurations
RQ1	3 models; `mixed_real`, `chat_multiturn`; $C \in \{16, 256\}$ (256 only for Qwen1.5); $S \in \{64, 256\}$; $\phi = \varnothing$
RQ2	Mistral-NeMo; `chat_multiturn`, `chat_real_long`; $C = S = 256$; $\phi \in \{\varnothing, \{\text{PREFIX}\}, \{\text{CHUNK}(512)\}, \{\text{PREFIX}, \text{CHUNK}(512)\}\}$
RQ3	Mistral-NeMo; `mixed_real`, `chat_multiturn`; $C = 256$; $S \in \{32, 64\}$; $\phi \in \{\varnothing, \{\text{SPEC}(3)\}\}$ (n-gram)
RQ4	**RQ1** protocol on GB10 with $C = 256$

`chat_multiturn` replays ShareGPT conversations of two to six turns. `chat_real_long` uses single-turn ShareGPT prompts of at least 500 tokens; prompts longer than the context (1%) are rejected by the client and excluded.

4.4 Load and measurement parameters

Arrival processes are Poisson with $\lambda \in \{2, 4, 8, 16\}$ requests/s and bursty with $B = 10$, $\delta = 50$ ms, $T_B = 5$ s. The client concurrency limit is $C \in \{16, 256\}$ and the scheduler limit $S \in \{64, 256\}$. Timeouts are $\tau_1 = 300$ s and $\tau_2 = 600$ s. Engine metrics are polled every Δt [VERIFY:] seconds. We evaluate goodput (Equation (11)) at a *strict* target $(T^\star, P^\star) = (2\,\text{s}, 200\,\text{ms})$ and a *relaxed* target $(4\,\text{s}, 500\,\text{ms})$, representative of interactive chat. [TODO: state warm-up policy and inter-run cool-down]

4.5 Experimental design

Table 4 lists the experiments per research question. Every configuration is run with all three engines and all five arrival processes.

In total the study comprises [TODO: count] runs and [TODO: count] requests on the L40S, and 180 runs and 72,000 requests on the GB10. Each configuration is run once.

5 RESULTS

5.1 RQ1: Engine behaviour under load

[TODO: write from Flow C]

5.2 RQ2: Prefill handling

[TODO:]

5.3 RQ3: Speculative decoding

[TODO:]

Table 5: Engine choice by deployment situation.

Situation	Prefer	Basis
Uniform requests, steady load	TensorRT-LLM	**RQ1**
Mixed lengths, TTFT-bound SLO	vLLM	**RQ1**
Smooth streaming required	SGLang/ TensorRT-LLM	**RQ1**
Model barely fits in KV memory	vLLM, more memory	**RQ1**, **RQ4**
Slot-bound deployment	raise `max_num_seqs`	**RQ1**
Long prompts	chunked prefill on TensorRT-LLM only	**RQ2**
Multi-turn chat	prefix caching on any engine	**RQ2**
N-gram speculation	vLLM only	**RQ3**

5.4 RQ4: Hardware generalisation

[TODO:]

6 DISCUSSION

6.1 Choosing an engine

Table 5 condenses the results into guidance. Its central message is that the right engine depends on three properties of the deployment: how heterogeneous the request lengths are, which latency the SLO is bound by, and how much KV-cache headroom the model leaves. [TODO: finalise the table once Section 5 is written]

6.2 Lessons for benchmarking LLM serving

Several measurement choices that look innocuous changed our conclusions. We report them because they apply to any LLM serving benchmark.

Measure TTFT from arrival, not from dispatch. A client that caps in-flight requests and starts the TTFT clock at dispatch silently excludes the time requests spend queued at the client—the classic coordinated-omission problem [27]. In our data this hidden wait reached a 95th percentile of over 200 s for the most overloaded engine, so dispatch-based TTFT understated the gap between engines precisely where it was largest. Open-loop arrival-based measurement [24] avoids this; for closed-loop multi-turn workloads, dispatch-based TTFT is the correct choice.

Count timeouts; do not drop them. Discarding requests that time out removes an overloaded engine's worst latencies and can invert rankings. Censoring them at the timeout value and counting them as SLO misses keeps the comparison honest.

Finite traces make latency non-monotonic in load. With a fixed number of requests, a low arrival rate stretches the trace over minutes, so late arrivals meet a KV cache already full of long-running generations, while at a high rate all requests arrive before the cache fills. In our data the 95th-percentile TTFT of one configuration was 185 s at 2 requests/s and 0.3 s at 16 requests/s; the cost moved from the first token to mid-stream stalls. Reporting latency against position in the run, or using duration-based traces, exposes this effect.

Identically named engine metrics are not comparable. Engines report prefix-cache hit rates over different windows (cumulative, last batch, or once per run), and the gap between streamed chunks shrinks or grows with the number of tokens per chunk under speculative decoding. We therefore compare engines only on client-side metrics and use engine-reported values to explain them. TPOT, unlike TBT, is robust to multi-token chunks.

Prompt formatting affects load, not just quality. Sending unformatted prompts to instruction-tuned models changed how many tokens they generated and, for some models, produced replies with zero tokens. Both change the load an engine experiences; a benchmark should either apply each model's chat template or report output-length distributions alongside latency.

Per-token speed is a poor proxy for service quality. Under the SLOs we study, goodput was decided almost entirely by TTFT: requests that missed only the TPOT target were at most 2% in every configuration. Benchmarks that report tokens per second at a comfortable load measure the property that least often limits real deployments.

7 THREATS TO VALIDITY

Construct validity. Our goodput SLOs (2 s/200 ms and 4 s/500 ms) are representative of interactive chat but not of every application; a tighter TPOT target favours the engine with the fastest decode, and we report that variant in the supplementary material. Engine-reported metrics have different semantics across engines (Section 3.3); we use them only to explain client-side results. Prefix-cache hit rates are measured per server, not per request.

Internal validity. Configuration options are mapped to each engine on a best-effort basis; "the same" nominal setting (e.g., chunk size, sequence limit, n-gram window) reflects each engine's own implementation. Two configuration defects were found and fixed before the corresponding sweeps (Section 4.1); others may remain in less-exercised code paths. Enabling n-gram speculation on TensorRT-LLM required disabling its overlap scheduler, which confounds that engine's speculative-decoding result [TODO: add a baseline with the overlap scheduler disabled]. TensorRT-LLM's KV-cache and queue statistics were not observable live, so its scheduler behaviour is inferred from client-side measurements. [TODO: confirm its capacity-scheduler policy from the launch log]

Prompts are not chat-templated for Llama-3.1-8B and Mistral-NeMo-12B. Llama-3.1-8B consequently generates about six times longer reasoning outputs than the other models, raising its KV pressure on `mixed_real`, and Mistral-NeMo-12B returns zero-token replies for 7–17% of requests on every engine. Both effects are identical across engines and therefore do not bias engine comparisons within a model, but they limit comparisons across models.

External validity. Results come from single-GPU deployments of 8–14B-parameter models on two platforms, with fp16 weights and no tensor parallelism. Larger models, quantisation, multi-GPU deployments, and disaggregated serving may change both absolute numbers and rankings. The **RQ4** platform also uses newer engine versions, so hardware and software effects are not separated

there. Arrival processes are synthetic, and real traffic may differ in burstiness and length mix.

Conclusion validity. Each configuration was run once, so we report no confidence intervals. We mitigate this by relying on effects that are large relative to run-to-run variation (often an order of magnitude), by observing them consistently across arrival rates and models, and by reproducing the main ranking on a second platform. [TODO: add repeated runs for a subset of configurations]

8 RELATED WORK

LLM serving systems. Orca introduced iteration-level scheduling [30]; vLLM added paged KV-cache management [12]; SGLang added prefix sharing through a radix-tree cache and a frontend language for structured programs [31]; TensorRT-LLM provides NVIDIA's optimised kernels and in-flight batching [19]. Research systems improve on specific aspects: chunked prefill to bound decode stalls [4], prefill–decode disaggregation [20, 32], preemptive and fair scheduling [29], live migration across instances [26], and KV-cache-centric architectures [21]. These papers evaluate their own design against baselines; we instead compare widely deployed engines as black boxes, through the interface applications use. A survey of serving techniques is given by Miao et al. [17].

Benchmarking and evaluating LLM inference. MLPerf Inference [22] defines scenarios and latency constraints for inference benchmarks but targets submitted, tuned systems rather than default deployments under realistic mixed traffic. Etalon [2] argues that TTFT, TBT and TPOT individually miss user-visible stalls and proposes a fluidity metric; Andes [15] defines quality of experience for streamed text. Vidur [3] simulates serving configurations to search for good deployments. Tools such as LLMPerf [6] measure latency and throughput of serving endpoints. Our study complements these with a controlled cross-engine comparison under saturation, an explanation of the differences in terms of admission and eviction behaviour, and a set of methodology pitfalls observed in practice.

Performance-engineering methodology. Open- and closed-loop load generators produce different conclusions about the same system [24], and latency recorders that measure from dispatch suffer from coordinated omission [27]. Little's law [14] relates queue length, arrival rate and residence time; we use it to explain when the scheduler's sequence limit becomes the bottleneck. Our methodology lessons (Section 6.2) apply these principles to LLM serving, where long, variable-length generations make them especially consequential.

[TODO: add recent empirical comparisons of serving engines (2025–2026) and position against them]

9 CONCLUSION

We compared vLLM, SGLang, and TensorRT-LLM under load across six user-facing metrics and SLO goodput, three models with different KV-cache footprints, realistic workloads, and two hardware platforms. The engines behave alike until a resource saturates; beyond that point they differ mainly in how they admit requests and where they make them wait, and those differences decide which engine meets a given SLO. No engine is best across workloads, metrics, and features: the fastest decoder can deliver the lowest goodput, and a feature that helps one engine can hurt another. We hope the decision guide and the methodology lessons help practitioners choose and evaluate serving engines for their own traffic. [TODO: finalise against the results]

Data and artifact availability. The benchmark harness, configuration files, per-request measurements for all runs, and the scripts that produce every figure and table will be released as an archived artifact with a DOI. [TODO: Zenodo DOI; link withheld for double-anonymous review]

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